

■ Nuclear Energy: Fission & Fusion

Grade 6 STEM Lesson | Engaging Science for You

■ Learning Objectives

By the end of this lesson you will be able to:

- Explain what nuclear energy is and where it comes from.
- Describe the structure of an atom (nucleus, protons, neutrons, electrons).
- Distinguish between nuclear fission and nuclear fusion.
- Give real-world examples of how each process is used.
- Compare the safety and environmental impacts of fission vs. fusion.

1. What is Nuclear Energy?

Nuclear energy is a special kind of power that comes from the very centre of atoms — called the **nucleus**. It is one of the most powerful energy sources we know. Scientists use this energy to generate electricity that powers our homes, schools, and cities. Energy is released whenever changes occur inside the tiny nucleus of an atom.

■ Fun Fact

A single kilogram of uranium fuel used in a nuclear power plant can produce as much electricity as burning roughly 3,000 tonnes of coal — with no carbon dioxide emissions!

2. Atoms — The Building Blocks

Everything around us is made of atoms — from the air we breathe to the food we eat. Each atom has three key parts:

Part	Location	Charge	Key Fact
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Proton	Nucleus	Positive (+)	Determines the element
Neutron	Nucleus	Neutral (0)	Adds mass; involved in fission
Electron	Orbiting shells	Negative (–)	Involved in chemical reactions

These building blocks are so tiny that millions of atoms could fit on the tip of a pencil!

3. Nuclear Fission — Splitting the Atom

Fission means splitting a large, heavy nucleus into smaller parts. Imagine a large atom like **Uranium-235**. When its nucleus is struck by a slow-moving neutron, something amazing happens — the nucleus breaks apart into smaller pieces, releasing a tremendous amount of energy and more neutrons. Think of it like breaking open a piñata: when it splits, everything bursts out!

How Fission Works — Step by Step:

Step 1: A neutron strikes a large nucleus (e.g. Uranium-235).

Step 2: The nucleus becomes unstable and splits into two smaller nuclei.

Step 3: Energy (heat and radiation) and two or three new neutrons are released.

Step 4: The released neutrons strike other uranium nuclei — a chain reaction begins!

Step 5: The chain reaction is carefully controlled inside a nuclear reactor to produce steady heat.

■ Uses of Nuclear Fission

- **Electricity generation:** Fission heats water → steam → turbines → electricity for millions of homes.
- **Medicine:** Nuclear energy is used to treat cancer and for imaging (e.g. PET scans).
- **Navy vessels:** Nuclear-powered ships and submarines can travel for years without refuelling.

4. Nuclear Fusion — Powering the Stars

Fusion means joining two small nuclei together to form a larger one. Imagine two hydrogen atoms coming together and combining into one helium atom. When this happens, an even *greater* amount of energy is released than fission! Fusion is the very process that powers our Sun and every star in the universe. Scientists around the world are working hard to harness fusion on Earth because it could provide virtually unlimited clean energy for our future.

How Fusion Works — Step by Step:

Step 1: Two small nuclei (e.g. hydrogen isotopes: deuterium and tritium) are brought very close together.

Step 2: Under extreme temperature (over 100 million °C!) and pressure, they overcome their repulsion.

Step 3: They fuse to form a larger nucleus (helium) plus a fast neutron.

Step 4: A massive amount of energy is released in the process.

■ **Uses & Potential of Fusion Energy**

- **Stars:** The Sun fuses ~620 million tonnes of hydrogen every second — giving us light and warmth.
- **Future reactors:** Projects like ITER (France) aim to produce net energy from fusion by the 2030s.
- **Clean power:** Fusion produces far less radioactive waste than fission, making it much safer for the planet.

5. Fission vs. Fusion — Side by Side

■ Nuclear Fission	■ Nuclear Fusion
• Splits a heavy nucleus into smaller parts	• Joins light nuclei to make a heavier one
• Used in today's nuclear power plants	• Powers the Sun and all stars
• Produces radioactive waste needing careful storage	• Much cleaner — very little radioactive waste
• Fuel: Uranium or Plutonium	• Fuel: Hydrogen isotopes (abundant in seawater!)
• Chain reaction can be controlled in a reactor	• Still experimental — scientists working hard on it

6. Safety & Environmental Impact

Nuclear power plants have many layers of safety systems to protect people and the environment:

- Thick concrete and steel containment structures surround the reactor core.
- Control rods absorb neutrons to slow down or stop the chain reaction.
- Multiple coolant systems prevent overheating.

- Radioactive waste from fission must be stored safely for thousands of years.

Fusion energy is far cleaner and safer — it produces much less radioactive waste and is why researchers around the world are excited about its potential as a future energy source!

7. The Future of Nuclear Energy

Nuclear energy — especially fusion — could play a major role in solving climate change. It produces no greenhouse gases and uses small amounts of fuel to generate huge amounts of electricity. Scientists are working hard to make fusion practical for everyday use. The most exciting part? **You** could be the future scientist who helps unlock this power! Your curiosity today could power tomorrow's world.

8. Check Your Understanding — Quiz

Q1. What is the centre of an atom called?

Answer: The nucleus, which contains protons and neutrons.

Q2. What is the difference between fission and fusion?

Answer: Fission splits a heavy nucleus into smaller parts; fusion joins small nuclei together.

Q3. Name ONE real-world use of nuclear fission.

Answer: Electricity generation in power plants / medical treatments / powering submarines.

Q4. Which process powers the Sun — fission or fusion?

Answer: Fusion. The Sun fuses hydrogen atoms into helium, releasing enormous energy.

Q5. Why might fusion energy be better for the environment than fission?

Answer: Fusion produces much less radioactive waste, making it cleaner and safer.

Q6. What is a 'chain reaction' in nuclear fission?

Answer: When neutrons released by one fission event trigger further fission events in neighbouring nuclei.

9. Key Vocabulary Glossary

Atom	The smallest unit of matter; made up of a nucleus and electrons.
Nucleus	The dense centre of an atom containing protons and neutrons.
Proton	A positively charged particle found in the nucleus.
Neutron	A neutral (no charge) particle found in the nucleus.
Electron	A negatively charged particle that orbits the nucleus.
Nuclear Fission	The splitting of a heavy nucleus into smaller nuclei, releasing energy.
Nuclear Fusion	The joining of two light nuclei into a heavier one, releasing energy.
Chain Reaction	A self-sustaining series of fission events triggered by released neutrons.
Radioactive Waste	By-products of fission that emit radiation and must be stored safely.
Uranium	A heavy radioactive element commonly used as fuel in fission reactors.
Turbine	A machine spun by steam to generate electricity in a power plant.

10. Extension Activity — Think About It!

■ Design a Fusion Reactor!

Imagine you are a scientist tasked with building a fusion power plant for your city. In your notebook, sketch and describe your reactor. Consider:

- What fuel will you use, and where will you get it from?
- How will you create temperatures of over 100 million °C?
- How will you capture the energy safely and convert it to electricity?
- What safety features will protect the surrounding community?

Share your design with a partner and discuss the challenges!